NATIONAL UNIVERSITY
COLLEGE OF COMPUTING AND
INFORMATION TECHNOLOGIES

VER 1.0

COURSE MANUAL

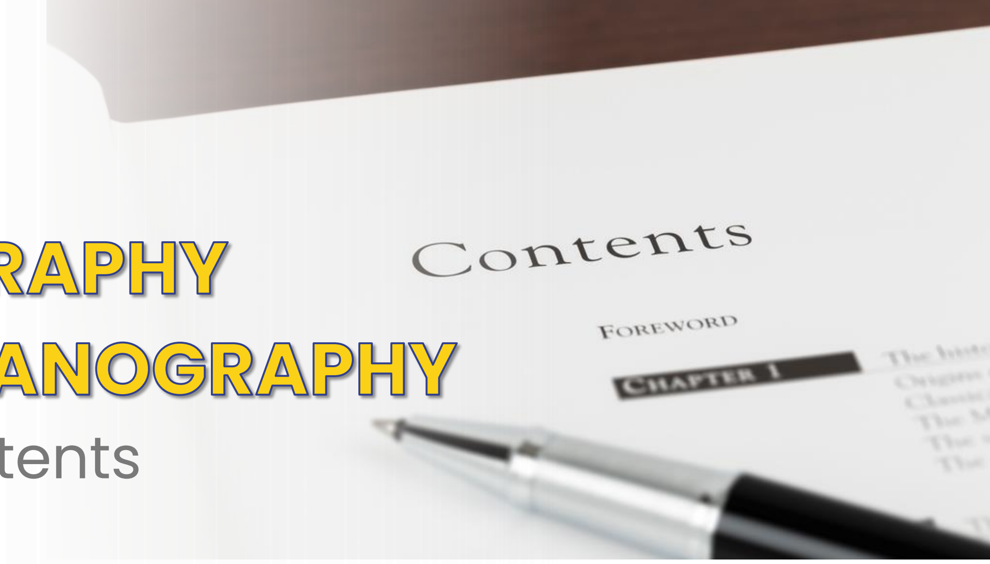
INFORMATION ASSURANCE AND SECURITY

This course provides students with the theoretical knowledge and practical skills in the implementation of Information Assurance and Security providing relevant solutions for security hardening and information protection through computer vulnerability assessment, secure coding, and computer security hardening techniques.

BS – COMPUTER SCICENCE
BS – INFORMATION TECHNOLOGY

Security and System Administration Cluster

MODULE 5
CRYPTOGRAPHY AND STEGANOGRAPHY



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Lesson Topics

Foundations of Cryptography

Learning Outcomes:

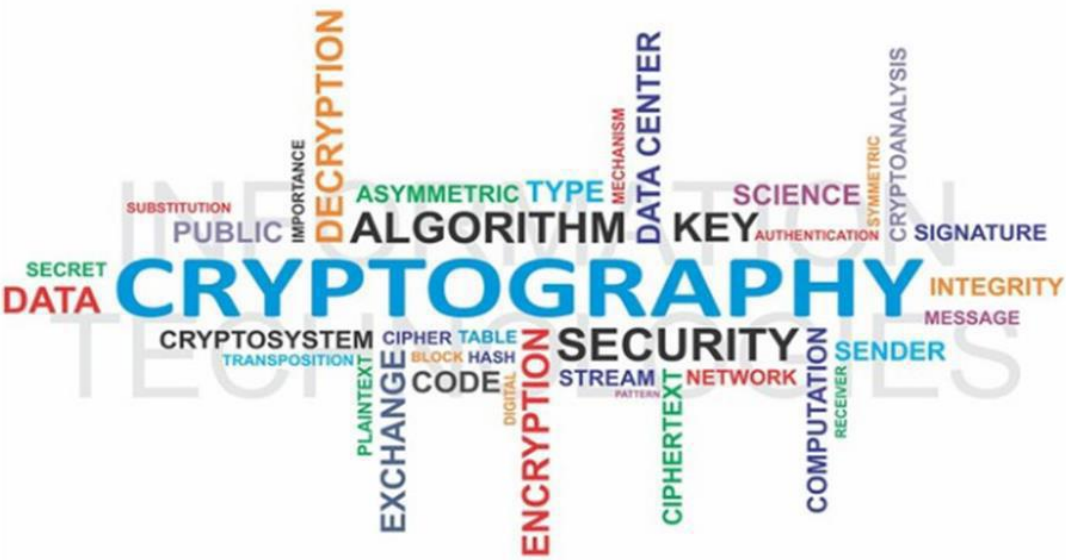
- Define cryptography and explain its role in securing digital information.
- Explain key terms such as plaintext, ciphertext, encryption, decryption, key, and algorithm.
- Understand the historical evolution of cryptography.

LESSON 1-1

Understanding Cryptography

The science of cryptography is not as enigmatic as you might think. A variety of cryptographic techniques are used regularly in everyday life. For example, open your newspaper to the entertainment section and you'll find the daily cryptogram, a word puzzle that involves unscrambling letters to find a hidden message. Also, although it is a dying art, many secretaries still use shorthand, or stenography, an abbreviated, symbolic writing method, to take rapid dictation.

The science of encryption, known as **cryptology**, encompasses cryptography and cryptanalysis. **Cryptography** comes from the Greek words *kryptos*, meaning "*hidden*," and *graphein*, meaning "*to write*," and involves making and using codes to secure messages. **Cryptanalysis** involves cracking or breaking encrypted messages back into their unencrypted origins. Cryptography uses mathematical algorithms that are usually known to all. After all, it's not the knowledge of the algorithm that protects the encrypted message, it's the knowledge of the key—a series of characters or bits injected into the algorithm along with the original message to create the encrypted message. An individual or system usually encrypts a plaintext message into ciphertext, making it unreadable to unauthorized people—those without the key needed to decrypt the message back into plaintext, where it can be read and understood.



History of Cryptology

Today, many common IT tools use embedded encryption technologies to protect sensitive information within applications. For example, all the popular Web browsers use built-in encryption features to enable secure ecommerce, such as online banking and Web shopping.

Since World War II, there have been restrictions on the export of cryptosystems, and they continue today. In 1992, encryption tools were officially listed as Auxiliary Military Technology under the Code of Federal Regulations: International Traffic in Arms Regulations. These restrictions are due in part to the role cryptography played in World War II, and the belief of the American and British governments that the cryptographic tools they developed were far superior to those in lesser developed countries. As a result, both governments believe such countries should be prevented from using cryptosystems to communicate potential terroristic activities or gain an economic advantage.

Terminology

- **Algorithm**: The steps used to convert an unencrypted message into an encrypted sequence of bits that represent the message; sometimes refers to the programs that enable the cryptographic processes.
- **Bit stream cipher:** An encryption method that involves converting plaintext to ciphertext one bit at a time.
- **Block cipher**: An encryption method that involves dividing the plaintext into blocks or sets of bits and then converting the plaintext to ciphertext one block at a time.
- **Cipher or cryptosystem:** An encryption method or process encompassing the algorithm, key(s) or cryptovariable(s), and procedures used to perform encryption and decryption.
- **Ciphertext or cryptogram:** The encoded message resulting from an encryption.

- **Code**: The process of converting components (words or phrases) of an unencrypted message into encrypted components.

- **Decipher/Decrypt:** To decrypt, decode, or convert ciphertext into the equivalent plaintext.

- **Encipher/Encrypt:** To encrypt, encode, or convert plaintext into the equivalent ciphertext.

- **Key or cryptovariable:** The information used in conjunction with an algorithm to create the ciphertext from the plaintext or derive the plaintext from the ciphertext. The key can be a series of bits used by a computer program, or it can be a passphrase used by people that is then converted into a series of bits used by a computer program.

- **Keyspace**: The entire range of values that can be used to construct an individual key

- **Plaintext or cleartext**: The original unencrypted message, or a message that has been successfully decrypted.

- **Steganography**: The hiding of messages—for example, within the digital encoding of a picture or graphic.

Cipher Methods

Learning Outcomes:

- Differentiate between classical cipher techniques, including substitution and transposition ciphers.
- Explain the concept of key space and brute-force attacks in the context of cipher strength.
- Familiarize tools used to encrypt and decipher messages.

Understanding Cipher Methods and Techniques

There are two methods of encrypting plaintext: the **bit stream method** or the **block cipher method.** In the bit stream method, each bit in the plaintext is transformed into a cipher bit one bit at a time. In the block cipher method, the message is divided into blocks—for example, sets of 8-, 16-, 32-, or 64-bit blocks—and then each block of plaintext bits is transformed into an encrypted block of cipher bits

using an algorithm and a key. Bit stream methods commonly use algorithm functions like the exclusive OR operation (XOR), whereas block methods can use substitution, transposition, XOR, or some combination of these operations.

Substitution Cipher

A **substitution cipher** exchanges one value for another—for example, it might exchange a letter in the alphabet with the letter three values to the right, or it might substitute one bit for another bit four places to its left. A three-character substitution to the right results in the following transformation of the standard English alphabet. Within this substitution scheme, the plaintext MOM would be encrypted into the ciphertext PRP.

Initial alphabet:	ABCDEFGHIJKLMNOPQRSTUVWXYZ	yields
Encryption alphabet:	DEFGHIJKLMNOPQRSTUVWXYZABC	

The previous example of substitution is based on a single alphabet and thus is known as a **monoalphabetic substitution**. More advanced substitution ciphers use two or more alphabets, and are referred to as **polyalphabetic substitutions**.

Plaintext:	ABCDEFGHIJKLMNOPQRSTUVWXYZ
Substitution cipher 1:	DEFGHIJKLMNOPQRSTUVWXYZABC
Substitution cipher 2:	GHIJKLMNOPQRSTUVWXYZABCDEF
Substitution cipher 3:	JKLMNOPQRSTUVWXYZABCDEFGHI
Substitution cipher 4:	MNOPQRSTUVWXYZABCDEFGHIJKL

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

The Vigenère Square

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An advanced type of substitution cipher that uses a simple polyalphabetic code is the Vigenère cipher. The cipher is implemented using the Vigenère square (or table), also known as a tabula recta—a term invented by Johannes Trithemius in the 1500s. Table above illustrates the setup of the Vigenère square, which is made up of 26 distinct cipher alphabets. In the header row and column, the alphabet is written in its normal order. In each subsequent row, the alphabet is shifted one letter to the right until a 26 x 26 block of letters is formed.

You can use the Vigenère square in several ways. For example, you could perform an encryption by simply starting in the first row, finding a substitute for the first letter of plaintext, and then moving down the rows for each subsequent letter of plaintext. With this method, the word SECURITY in plaintext becomes TGFYWOAG in ciphertext.

Transposition Cipher

Like the substitution operation, the transposition cipher is simple to understand, but if properly used, it can produce ciphertext that is difficult to decipher. In contrast to the substitution cipher, however, the **transposition cipher** or **permutation cipher** simply rearranges the bits or bytes (characters) within a block to create the ciphertext. For an example, consider the following transposition key pattern.

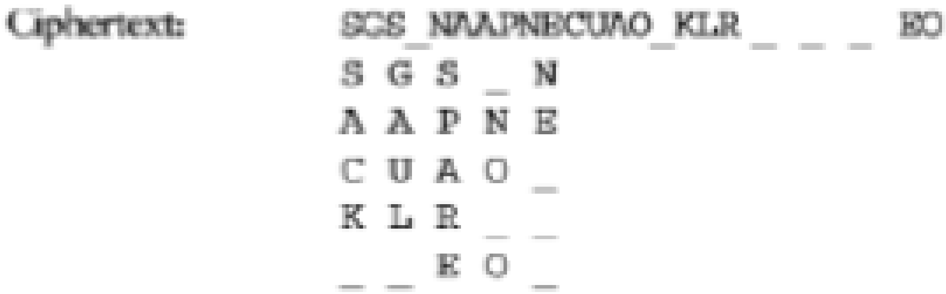
Key pattern: $8 \rightarrow 3, 7 \rightarrow 6, 6 \rightarrow 2, 5 \rightarrow 7, 4 \rightarrow 5, 3 \rightarrow 1, 2 \rightarrow 8, 1 \rightarrow 4$

In this key, the bit or byte (character) in position 1 moves to position 4. When operating on binary data, position 1 is at the far right of the data string, and counting proceeds from right to left. Next, the bit or byte in position 2 moves to position 8, and so on. To examine further how this transposition key works, look at its effects on a plaintext message comprised of letters. Replacing the 8-bit block of plaintext with the example plaintext message "SACK GAUL SPARE NO ONE," yields the following.

Letter locations:	87654321 | 87654321 | 87654321
Plaintext:	__ENO_ON | _ERAPS_L | UAG_KCAS
Key:	Same key as above, but characters transposed, not bits.
Ciphertext:	ON_ON_E_ | _AEPL_RS | A_AKSUGC

Here, you read from right to left to match the order in which characters would be transmitted from a sender on the left to a receiver on the right. The letter in position 1 of the first block of plaintext, "S," moves to position 4 in the ciphertext. The process is continued until the letter "U," the eighth letter of the first block of plaintext, moves to the third position of the ciphertext. This process continues with subsequent blocks using the same specified pattern. Obviously, the use of different-sized blocks or multiple transposition patterns would enhance the strength of the cipher.

In the Caesar block cipher, the recipient of the coded message knows to fit the text to a prime number square. In practice, this means that if there are fewer than 25 characters, the recipient uses a 5 × 5 square. For example, if you received the Caesar ciphertext shown below, you would make a square of five columns and five rows, and then write the letters of the message into the square, filling the slots from left to right and top to bottom. Then you would read the message from the opposite direction—that is, from top to bottom, left to right. Reading from top to bottom, left to right reveals the plaintext “SACK GAUL SPARE NO ONE.”

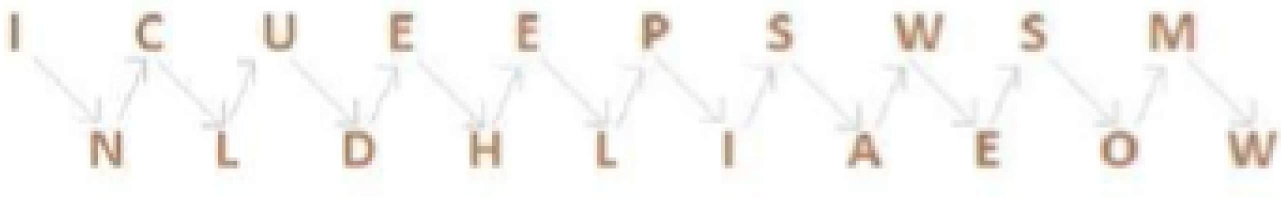


Transposition Cipher Techniques

Rail-Fence Technique

This technique is a type of Transposition technique and does is write the plain text as a sequence of diagonals and changing the order according to each row. It uses a simple algorithm,

- Writing down the plaintext message into a sequence of diagonals.
- Row-wise writing the plain-text written from above step. Example, Let’s say, we take an example of “INCLUDEHELP IS AWESOME”.



So the Cipher-text are, **ICUEEPSWSMNLDHLIAEOW.**

Columnar Transition Technique

It is a slight variation to the Rail-fence technique, let’s see its algorithm:

- In a rectangle of pre-defined size, write the plain-text message row by row.
- Read the plain message in random order in a column-wise fashion. It can be any order such as 2, 1, 3 etc.
- Thus Cipher-text is obtained.

Let’s see an example:

Original message: "INCLUDEHELP IS AWESOME". Now we apply the above algorithm and create the rectangle of 4 columns (we decide to make a rectangle with four column it can be any number.)

Column 1	Column 2	Column 3	Column 4
I	N	C	L
U	D	E	H
E	L	P	I
S	A	W	E
S	O	M	E

Now let’s decide on an order for the column as 4, 1, 3 and 2 and now we will read the text in column-wise.

Cipher-text: **LHIEEIUESSCEPWMNDLAO**

Vernam Cipher (One-Time Pad)
The Vernam Cipher has a specific subset one-time pad, which uses input ciphertext as a random set of nonrepeating character. The thing to notice here is that, once an input cipher text gets used it will never be used again hence one-time pad and length of cipher-text is the size that of message text.

Algorithm:

- Plain text character will be represented by the numbers as A=0, B=1, C=2,... Z=25.
- Add each corresponding number of a plain text message to the input cipher text alphabet numbers.
- If the sum is greater than or equal to 26, subtract 26 from it.
- Translate each number back to corresponding letters and we got our cipher text.

Example: Our message is "**INCLUDEHELP**" and input cipher text is **"ATQXRZWOBYV"**

	I	N	C	L	U	D	E	H	E	L	P
Plain text:	8	13	2	11	20	3	4	7	4	11	15
One-time pad:	0	19	16	23	17	25	22	14	1	24	21
	A	T	Q	X	R	Z	W	O	B	Y	V
Initial Total:	8	32	18	34	37	28	26	21	5	35	36
Subtract 26, if >25:	8	6	18	8	11	2	0	21	5	9	10
Cipher Text:	I	G	S	I	L	C	A	V	F	J	K

One time pad should be discarded after every single use and this technique is proved highly secure and suitable for small messages but illogical if used for long messages.

Book/Running-Key Cipher
This technique also (incorrectly) known as running key cipher. This technique very simple and similar to our previous Vernam Cipher. For getting a cipher, some portion of text from a book is used as a one-time pad, rest it works in same way as Vernam cipher does.

Hash Functions

In addition to ciphers, another important encryption technique that is often incorporated into cryptosystems is the hash function. **Hash functions** are mathematical algorithms used to confirm the identity of a specific message and confirm that the content has not been changed. While they do not create ciphertext, hash functions confirm message identity and integrity, both of which are critical functions in e-commerce.

Hash algorithms are used to create a hash value, also known as a message digest, by converting variable length messages into a single fixed-length value. The message digest is a fingerprint of the author's message that is compared with the recipient's locally calculated hash of the same message. If both hashes are identical after transmission, the message has arrived without modification. Hash functions are considered one-way operations in that the same message always provides the same hash value, but the hash value itself cannot be used to determine the contents of the message.

Hashing functions do not require the use of keys, but it is possible to attach a **message authentication code (MAC)** to allow only specific recipients to access the message digest. Because hash functions are one-way, they are used in password verification systems to confirm the identity of the user. In such systems, the hash value, or message digest, is calculated based on the originally issued password, and this message digest is stored for later comparison. When the user logs on for the next session, the system calculates a hash value based on the user's password input, and this value is compared against the stored value to confirm identity.

The **Secure Hash Standard (SHS)** is issued by the National Institute of Standards and Technology (NIST). Standard document FIPS 180-4 specifies SHA-1 (Secure Hash Algorithm 1) as a secure algorithm for computing a condensed representation of a message or data file. SHA-1 produces a 160-bit message digest, which can be used as an input to a digital signature algorithm.

Various hash values

Source: SlavaSoft HashCalc

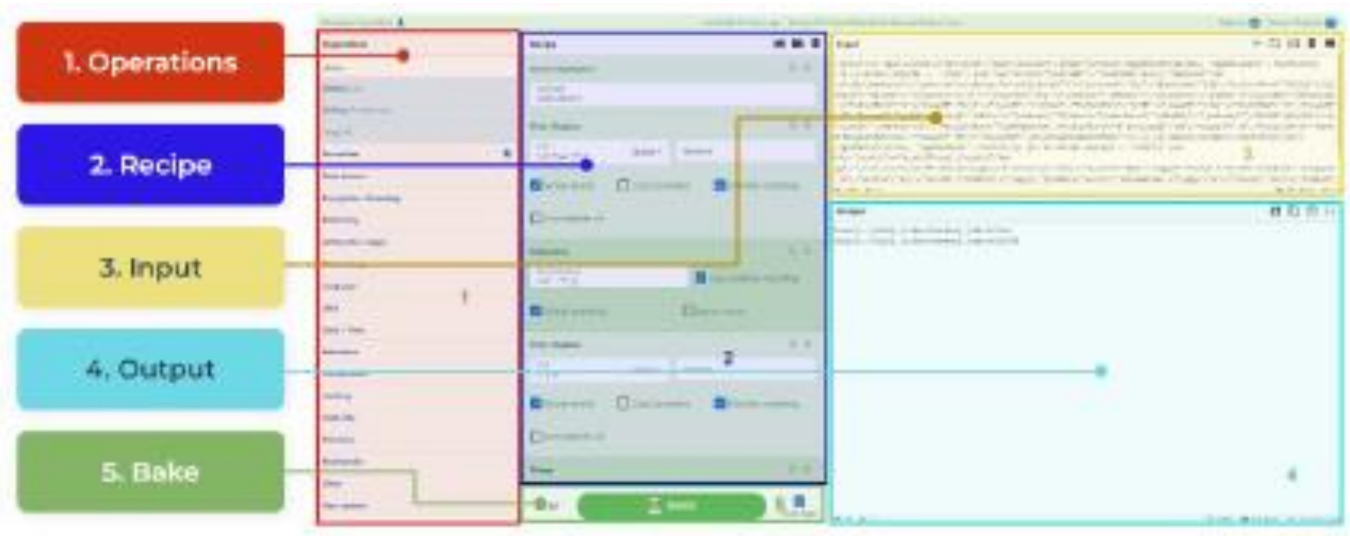
New hash algorithms, SHA-256, SHA-384, and SHA-512, have been proposed by NIST as standards for 128, 192, and 256 bits, respectively. The number of bits used in the hash algorithm is a measurement of the algorithm's strength against collision attacks. SHA-256 is essentially a 256-bit block cipher algorithm that creates a key by encrypting the intermediate hash value, with the message block functioning as the key.

Cipher Encoding and Decoding Tools

CyberChef - The Cyber Swiss Army Knife - is a web-based utility that allows analysts to manipulate or transform inputs based on a series of steps called a recipe. The versatile tool is used by a wide range of individuals, including cybersecurity analysts, researchers, and enthusiasts. Each step is an operation that takes place on the input and generates the corresponding output. The product was created by the GCHQ, subsequently open-sourced and available on GitHub.

The utility supports operations in the following areas:

- Encryption
- Encoding
- Networking
- Extracting Data
- Compression



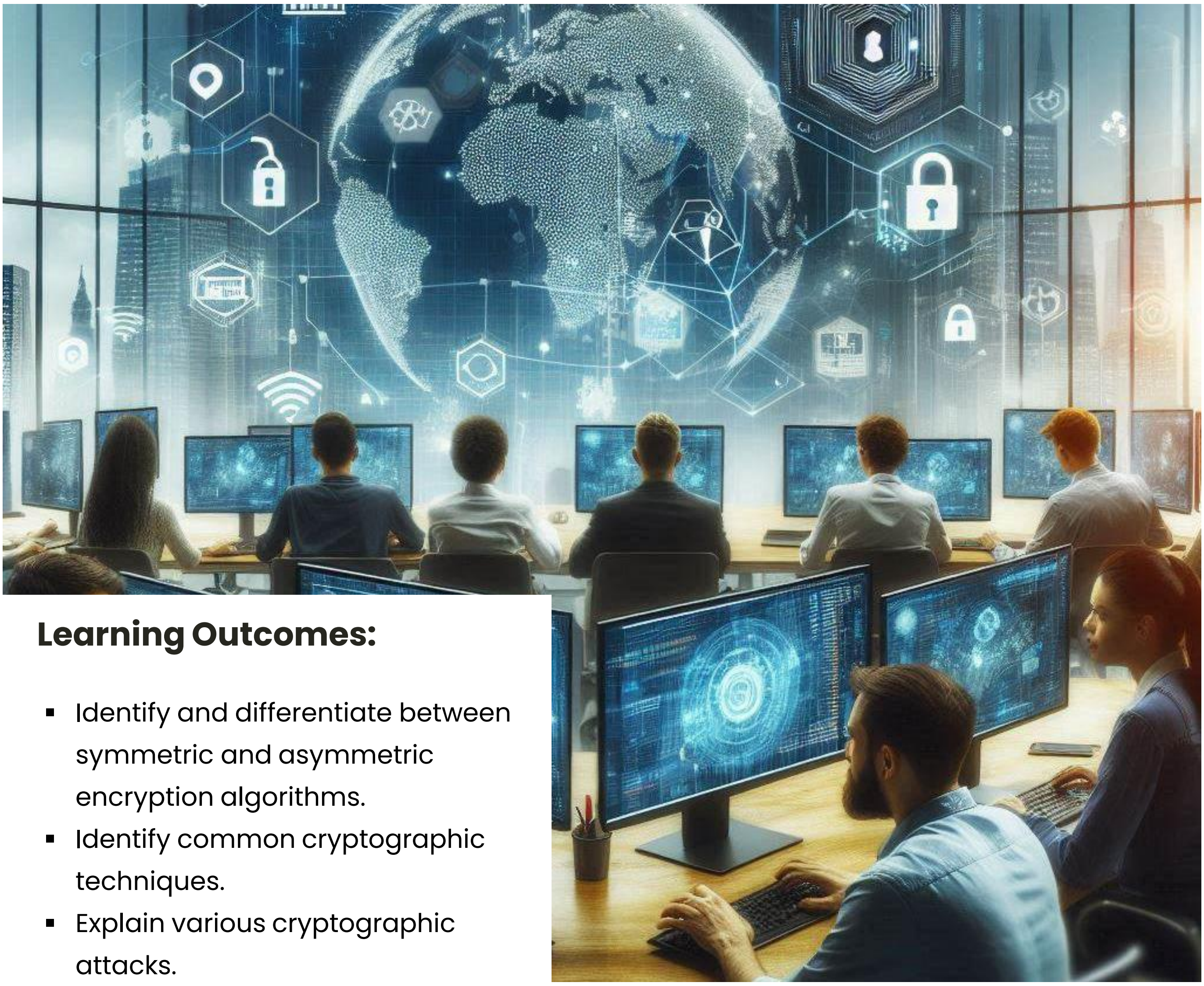
Online Tool -> https://gchq.github.io/CyberChef/

dCode is a toolkit website for decryption, ciphertexts, cheating at letter games, solve riddles, treasure hunts, etc. Opened in 2009, the site has grown steadily and improved, offering more and more tools and carried by its users, dCode has become a public site, free and open to all!

Online Tool -> https://www.dcode.fr/en

LESSON 3

Cryptographic Algorithms, Techniques and Attacks



Learning Outcomes:

- Identify and differentiate between symmetric and asymmetric encryption algorithms.
- Identify common cryptographic techniques.
- Explain various cryptographic attacks.

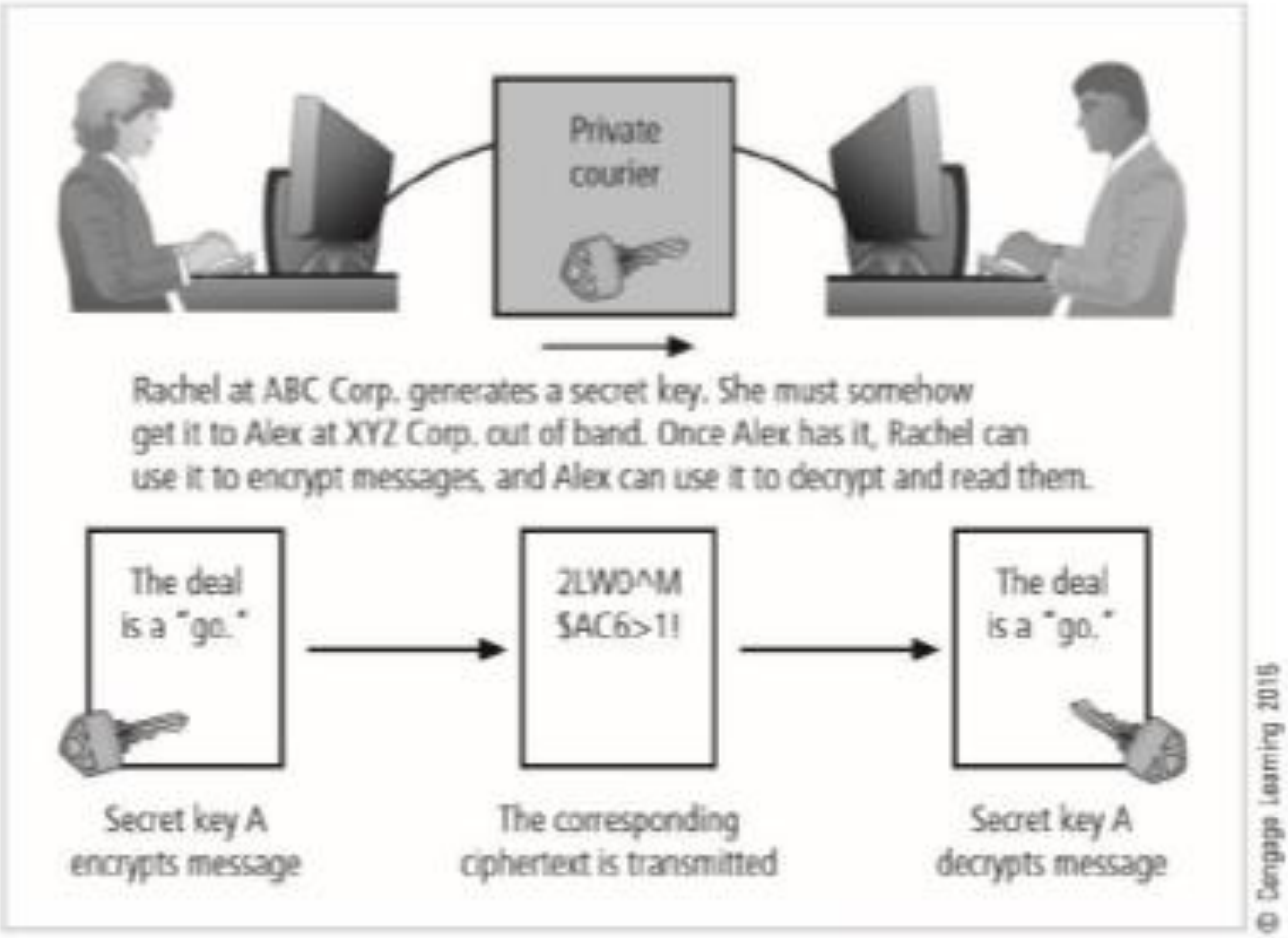
Understanding Cryptographic Algorithms

In general, cryptographic algorithms are often grouped into two broad categories—symmetric and asymmetric—but in practice, today's popular cryptosystems use a combination of both algorithms. Symmetric and asymmetric algorithms are distinguished by the types of keys they use for encryption and decryption operations.

Symmetric Encryption

Encryption methodologies that require the same secret key to encipher and decipher the message are performing private-key encryption or symmetric encryption. **Symmetric encryption** methods use mathematical operations that can be programmed into extremely fast computing algorithms so that encryption and decryption are executed quickly, even by small computers.

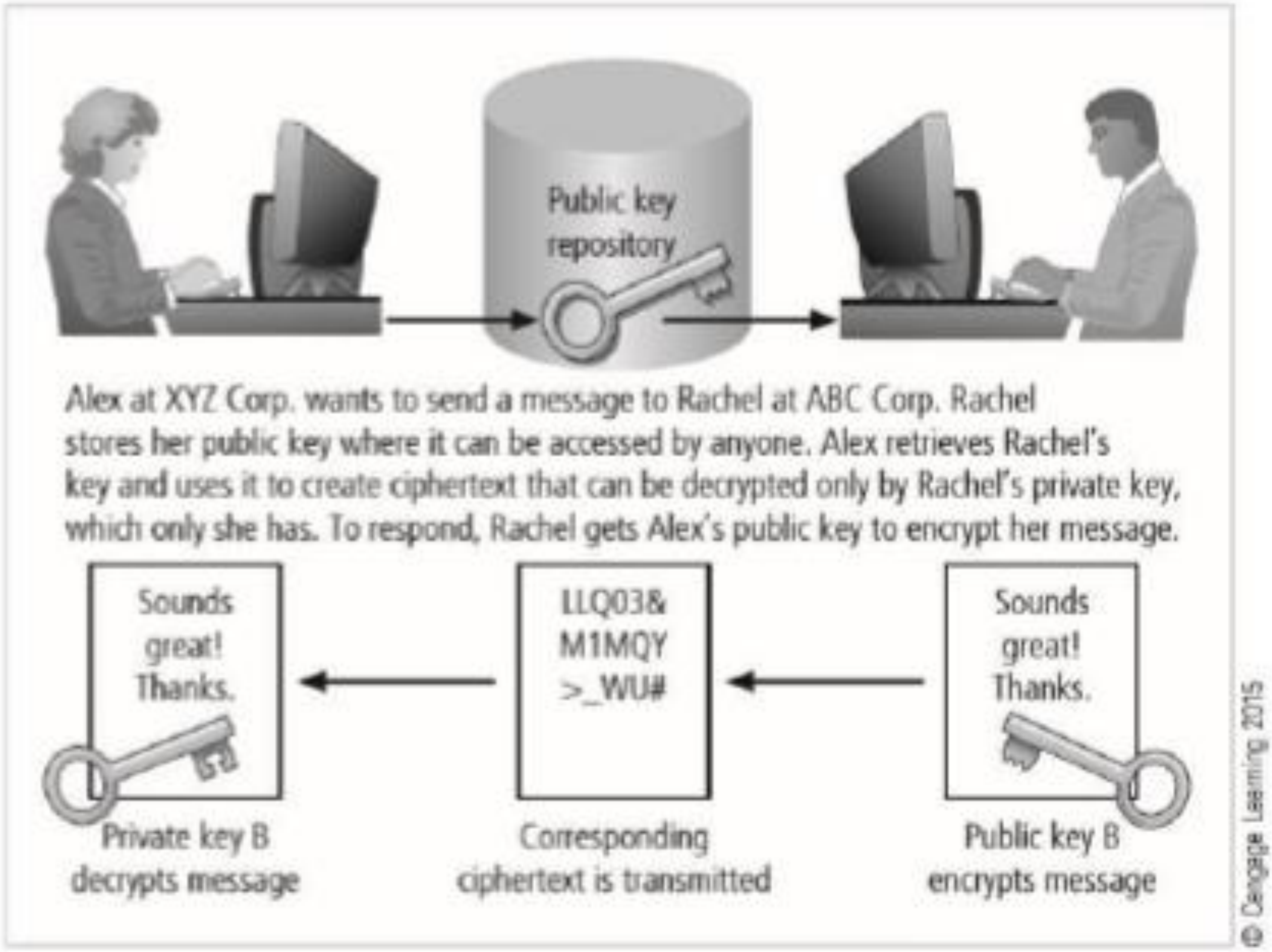
One of the challenges is that both the sender and the recipient must have the secret key. Also, if either copy of the key falls into the wrong hands, messages can be decrypted by others and the sender and intended receiver may not know a message was intercepted. The primary challenge of symmetric key encryption is getting the key to the receiver, a process that must be conducted out of band to avoid interception. In other words, the process must use a channel or band other than the one carrying the ciphertext.



Example of symmetric encryption

Asymmetric Encryption

While symmetric encryption systems use a single key both to encrypt and decrypt a message, **asymmetric encryption** (An encryption method that incorporates mathematical operations involving both a public key and a private key to encipher or decipher a message. Either key can be used to encrypt a message, but then the other key is required to decrypt it.) uses two different but related keys. Either key can be used to encrypt or decrypt the message. However, if key A is used to encrypt the message, only key B can decrypt it; if key B is used to encrypt a message, only key A can decrypt it. Asymmetric encryption can be used to provide elegant solutions to problems of secrecy and verification. This technique has its greatest value when one key is used as a private key, which means it is kept secret (much like the key in symmetric encryption) and is known only to the owner of the key pair. The other key serves as a public key, which means it is stored in a public location where anyone can use it. For this reason, the more common name for asymmetric encryption is public-key encryption.



Example of asymmetric encryption

Asymmetric algorithms are one-way functions, meaning they are simple to compute in one direction, but complex to compute in the opposite direction. This is the foundation of public-key encryption. It is based on a hash value, which is calculated from an input number using a hashing algorithm, as you learned earlier in this chapter. This hash value is essentially a summary of the original input values. It is virtually impossible to derive the original values without knowing how they were used to create the hash value.

Asymmetric Encryption

When deploying ciphers, it is important for users to decide on the size of the cryptovariable or key, because the strength of many encryption applications and cryptosystems is measured by key size. How exactly does key size affect the strength of an algorithm? Typically, the length of the key increases the number of random guesses that have to be made in order to break the code. Creating a larger universe of possibilities increases the time required to make guesses, and thus a longer key directly influences the strength of the encryption.

It may surprise you to learn that when it comes to cryptosystems, the security of encrypted data is not dependent on keeping the encrypting algorithm secret. In fact, algorithms should be published and often are, to enable research to uncover their weaknesses. The security of any cryptosystem depends on keeping some or all elements of the cryptovariable(s) or key(s) secret, and effective security is maintained by manipulating the size (bit length) of the keys and following proper procedures and policies for key management.

It is estimated that to crack an encryption key using a brute force attack, a computer needs to perform a maximum of 2^k operations (2^k guesses), where k is the number of bits in the key. In reality, the average estimated time to crack is half that time.			
Using an average 2013-era Intel i7 PC (3770K) chip performing 109,924 Dhrystone MIPS (million instructions per second) at 3.9 GHz:			
Key length (bits)	**Maximum number of operations (guesses)**	**Maximum time to crack**	**Estimated average time to crack**
16	65,636	0.00000061 seconds	0.00000031 seconds
24	16,777,216	0.00016 seconds	0.00008 seconds
32	4,294,967,296	0.04 seconds	0.02 seconds
56	72,057,594,037,927,900	7.8 days	3.9 days
64	18,446,744,073,709,600,000	5.48 years	2.74 years
128	3.40E+38	101,123,123,702,677, 000,000 years	50,561,561,851,038, 500,000 years
256	1.16E+77	34,410,426,468,960, 700,000,000,000,000, 000,000,000,000,000, 000,000,000,000,000 years	17,205,213,234,480,300, 000,000,000,000,000, 000,000,000,000,000, 000,000,000,000 years
512	1.34E+154	3,984,515,321,402,380, 000,000,000,000,000, 000,000,000,000,000, 000,000,000,000,000, 000,000,000,000,000, 000,000,000,000,000, 000,000,000,000,000, 000,000,000,000,000, 000,000,000,000,000, 000 years	1,992,257,660,701,190, 000,000,000,000,000, 000,000,000,000,000, 000,000,000,000,000, 000,000,000,000,000, 000,000,000,000,000, 000,000,000,000,000, 000,000,000,000,000, 000,000,000,000,000, 000 years

Encryption Key Power

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Understanding Cryptographic Techniques

The ability to conceal the contents of sensitive messages and verify the contents of messages and the identities of their senders can be important in all areas of business. To be useful, these cryptographic capabilities must be embodied in tools that allow IT and information security practitioners to apply the elements of cryptography in the everyday world of computing. This section covers some of the widely used tools that bring the functions of cryptography to the world of information systems.

Public Key Infrastructure (PKI)

Public key infrastructure (PKI) systems are based on public-key cryptosystems and include digital certificates and certificate authorities (CAs). Digital certificates allow the PKI components and their users to validate keys and identify key owners. (Digital certificates are explained in more detail later in this chapter.) PKI systems and their digital certificate registries enable the protection of information assets by making verifiable digital certificates readily available to business applications.

Common implementations of PKI include systems that issue digital certificates to users and servers, directory enrollment, key issuing systems, tools for managing key issuance, and verification and return of certificates. These systems enable organizations to apply an enterprise wide solution that allows users within the PKI's area of authority to engage in authenticated and secure communications and transactions.

The strength of a cryptosystem relies on both the raw strength of its key's complexity and the overall quality of its key management security. PKI solutions can provide several mechanisms for limiting access and possible exposure of the private keys. These mechanisms include password protection, smart cards, hardware tokens, and other hardware-based key storage devices that are memory-capable, like flash memory or PC memory cards. PKI users should select the key security mechanisms that provide an appropriate level of key protection for their needs. Managing the security and integrity of the private keys used for nonrepudiation or the encryption of data files is critical to successfully using the encryption and nonrepudiation services within the PKI's area of trust.

Digital Signatures

Digital signatures were created in response to the rising need to verify information transferred via electronic systems. Asymmetric encryption processes are used to create digital signatures. When an asymmetric cryptographic process uses the sender's private key to encrypt a message, the sender's public key must be used to decrypt the message. When the decryption is successful, the process verifies that the message was sent by the sender and thus cannot be refuted. This process is known as nonrepudiation, and is the principle of cryptography that underpins the authentication mechanism collectively known as a digital signature. **Digital signatures**, therefore, are encrypted messages that can be mathematically proven as authentic.

Digital signature in Windows 7 Internet Explorer

Source: Windows 7 Internet Explorer

The management of digital signatures is built into most Web browsers. For example, the digital signature management screen in Internet Explorer is shown in below. In general, digital signatures should be created using processes and products that are based on the Digital Signature Standard (DSS).

Digital Certificates

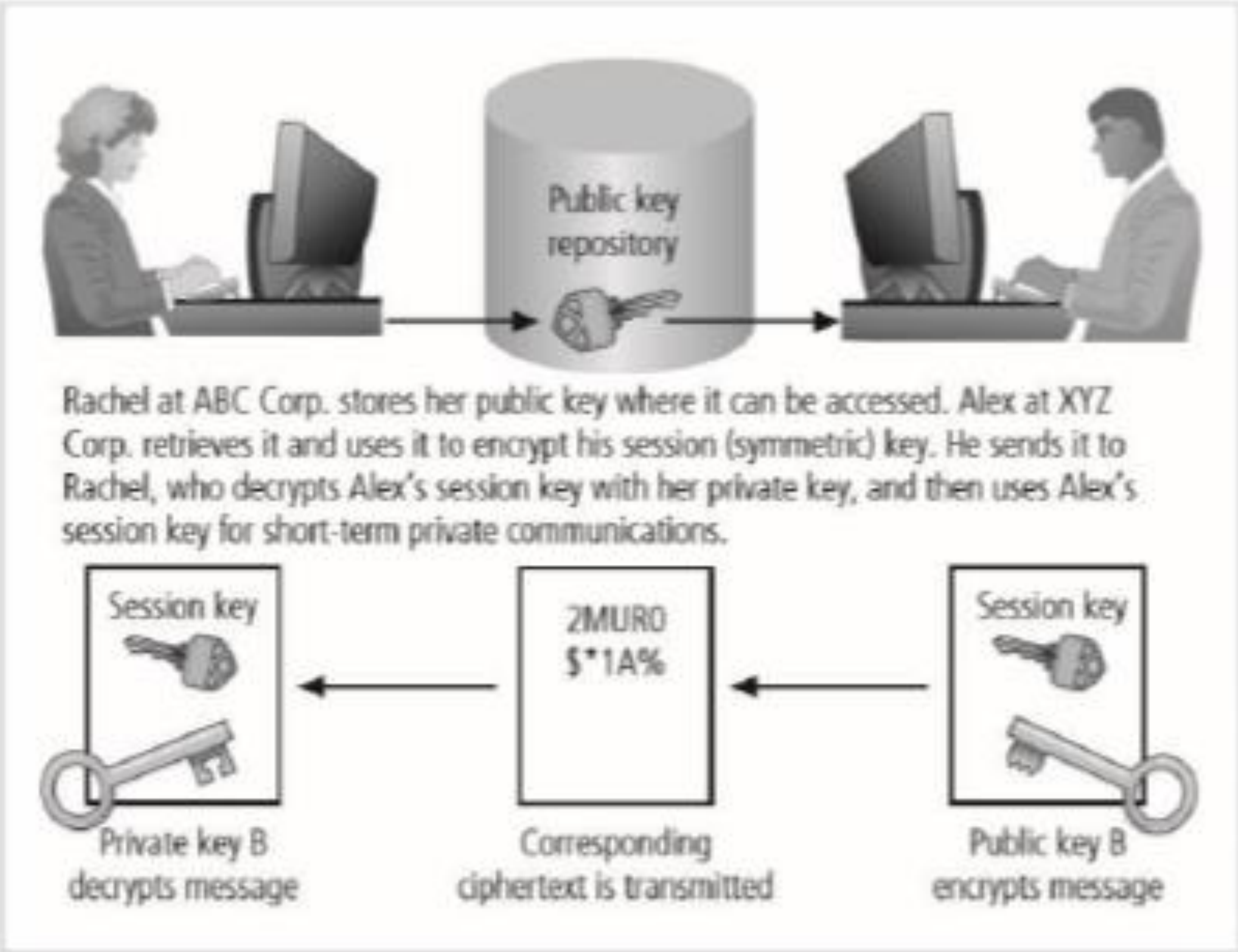
A digital certificate is an electronic document or container file that contains a key value and identifying information about the entity that controls the key. The certificate is often issued and certified by a third party, usually a certificate authority. A digital signature attached to the certificate's container file certifies the file's origin and integrity. This verification process often occurs when you download or update software via the Internet.

Example digital certificate

Source: Amazon.com

Hybrid Cryptography Systems

The most common hybrid system is based on the Diffie-Hellman key exchange, which uses asymmetric encryption to exchange session keys. These are limited-use symmetric keys that allow two entities to conduct quick, efficient, secure communications based on symmetric encryption, which is more efficient than asymmetric encryption for sending messages. Diffie-Hellman provides the foundation for subsequent developments in public key encryption. It protects data from exposure to third parties, which is sometimes a problem when keys are exchanged out of band.



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Example of hybrid encryption

Understanding Cryptography Attacks

A **cryptographic attack** is a method used by hackers to target cryptographic solutions like ciphertext, encryption keys, etc. These attacks aim to retrieve the plaintext from the ciphertext or decode the encrypted data. Hackers may attempt to bypass the security of a cryptographic system by discovering weaknesses and flaws in cryptography techniques, cryptographic protocol, encryption algorithms, or key management strategy.

Hybrid Cryptography Systems

Depending on the type of cryptographic system in place and the information available to the attacker, these attacks can be broadly classified into six types:

BRUTE FORCE ATTACK
Public and private keys play a significant role in encrypting and decrypting the data in a cryptographic system. In a brute force attack, the cybercriminal tries various private keys to decipher an encrypted message or data. If the key size is 8-bit, the possible keys will be 256 (i.e., 28). The cybercriminal must know the algorithm (usually found as open-source programs) to try all the 256 possible keys in this attack technique.

CIPHERTEXT-ONLY ATTACK
In this attack vector, the attacker gains access to a collection of ciphertext. Although the attacker cannot access the plaintext, they can successfully determine the ciphertext from the collection. Through this attack technique, the attacker can occasionally determine the key.

CHOSEN PLAINTEXT ATTACK
In this attack model, the cybercriminal can choose arbitrary plaintext data to obtain the ciphertext. It simplifies the attacker's task of resolving the encryption key. One well-known example of this type of attack is the differential cryptanalysis performed on block ciphers.

CHOSEN CIPHERTEXT ATTACK

In this attack model, the cybercriminal analyzes a chosen ciphertext corresponding to its plaintext. The attacker tries to obtain a secret key or the details about the system. By analyzing the chosen ciphertext and relating it to the plaintext, the attacker attempts to guess the key. Older versions of RSA encryption were prone to this attack.

KNOWN PLAINTEXT ATTACK

In this attack technique, the cybercriminal finds or knows the plaintext of some portions of the ciphertext using information gathering techniques. Linear cryptanalysis in block cipher is one such example.

KEY AND ALGORITHM ATTACK

Here, the attacker tries to recover the key used to encrypt or decrypt the data by analyzing the cryptographic algorithm.

Preventing Cryptography Attacks

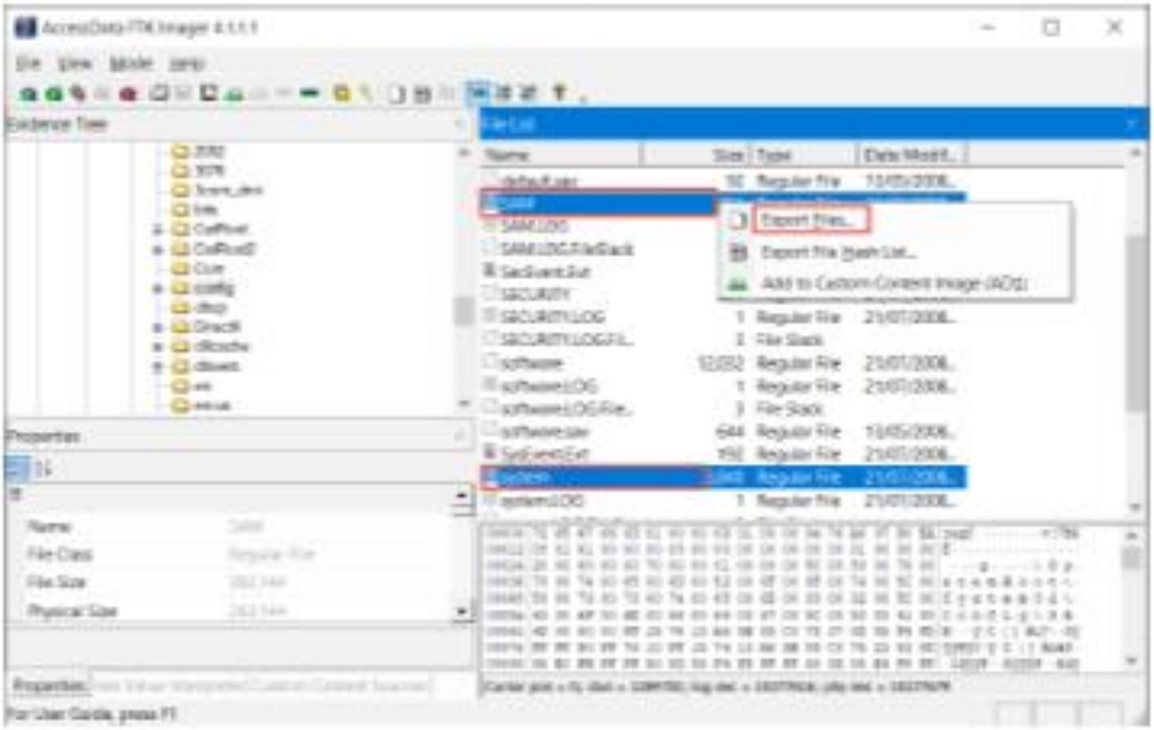
To prevent cryptography attacks, it is essential to have a strong cryptographic system in place. Some of the ways to achieve this are:

- Regularly update the cryptographic algorithms and protocols to ensure they are not obsolete.
- Ensure that the data is appropriately encrypted so that even if it falls into the wrong hands, it will be unreadable.
- Use strong and unique keys for encryption.
- Store the keys in a secure location.
- Ensure that the cryptographic system is implemented correctly.
- Regularly test the system for vulnerabilities.
- Educate employees about cryptography attacks and how to prevent them.

Example of Cryptography Attacks

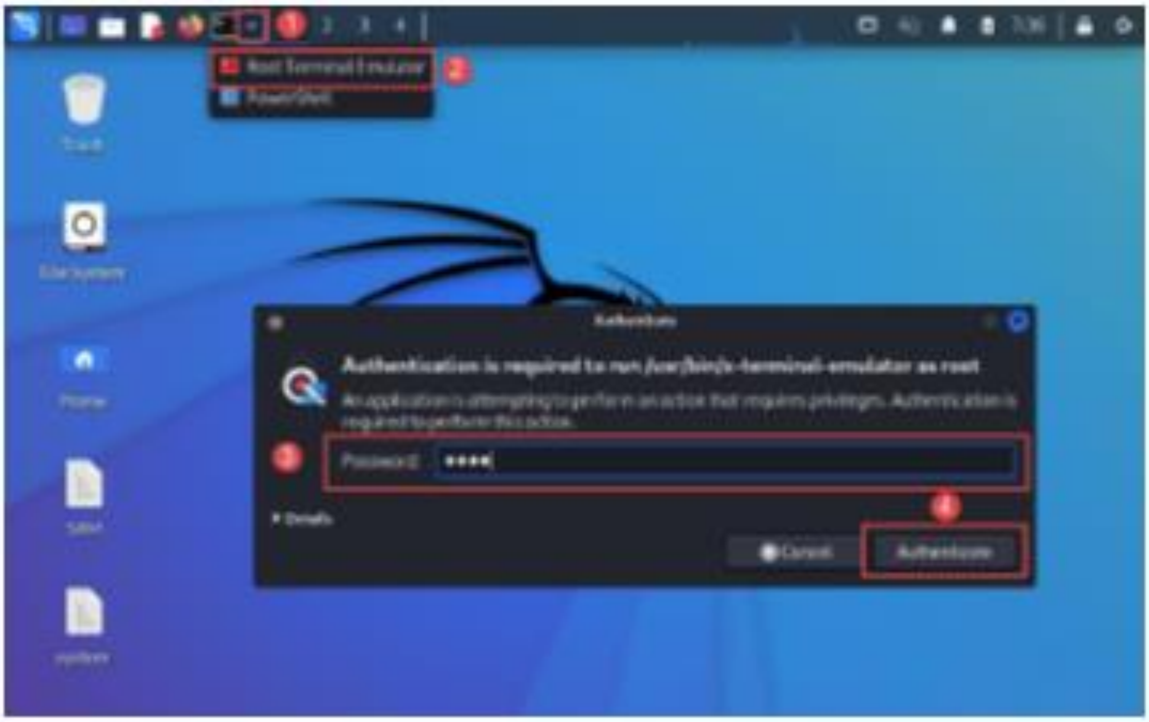
Crack Windows Passwords using Registry Hives

Step 1: Obtain the registry hive of interest (SAM and SYSTEM) using AccessData FTK Imager.



Step 2: Boot into your Kali workstation and transfer the obtained registry into Kali's Desktop as shown below:

Step 3: Open terminal window (as root), an authentication prompt will be displayed. Input the password "kali" and click on authenticate button to proceed.

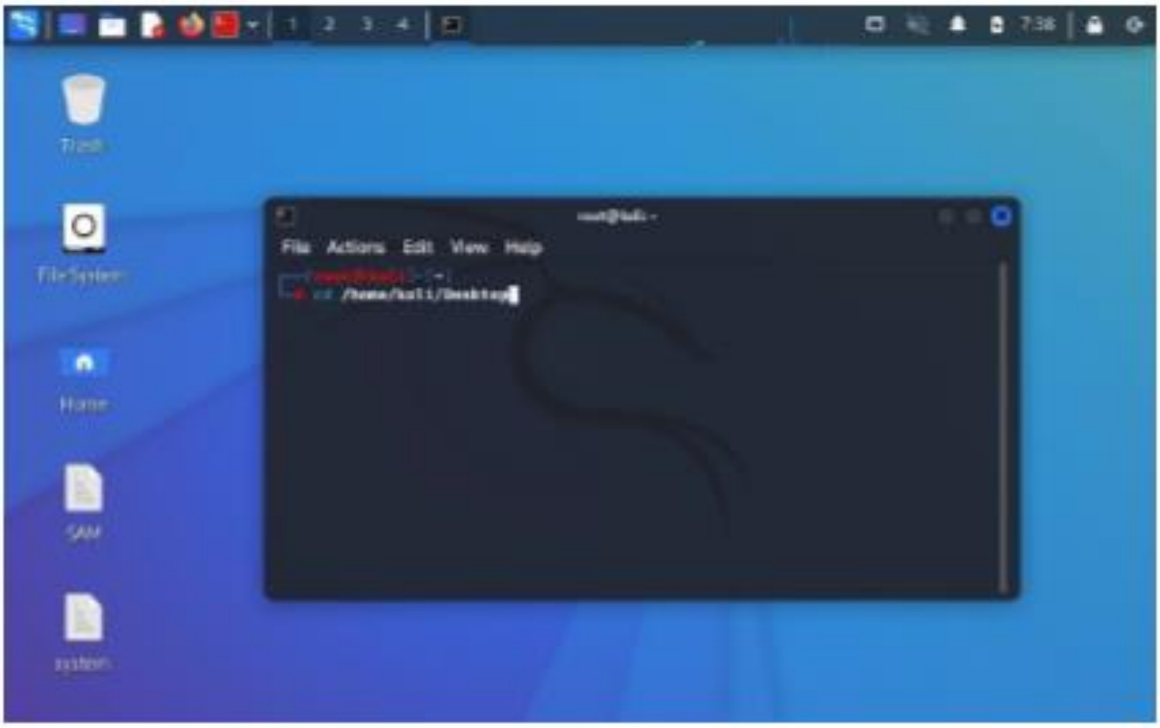


Step 4: Using the terminal window, update “John-the-Ripper” program.

Type Command:
apt-get install john

Step 5: Using the terminal window, Navigate to the directory where the registry hives are stored.

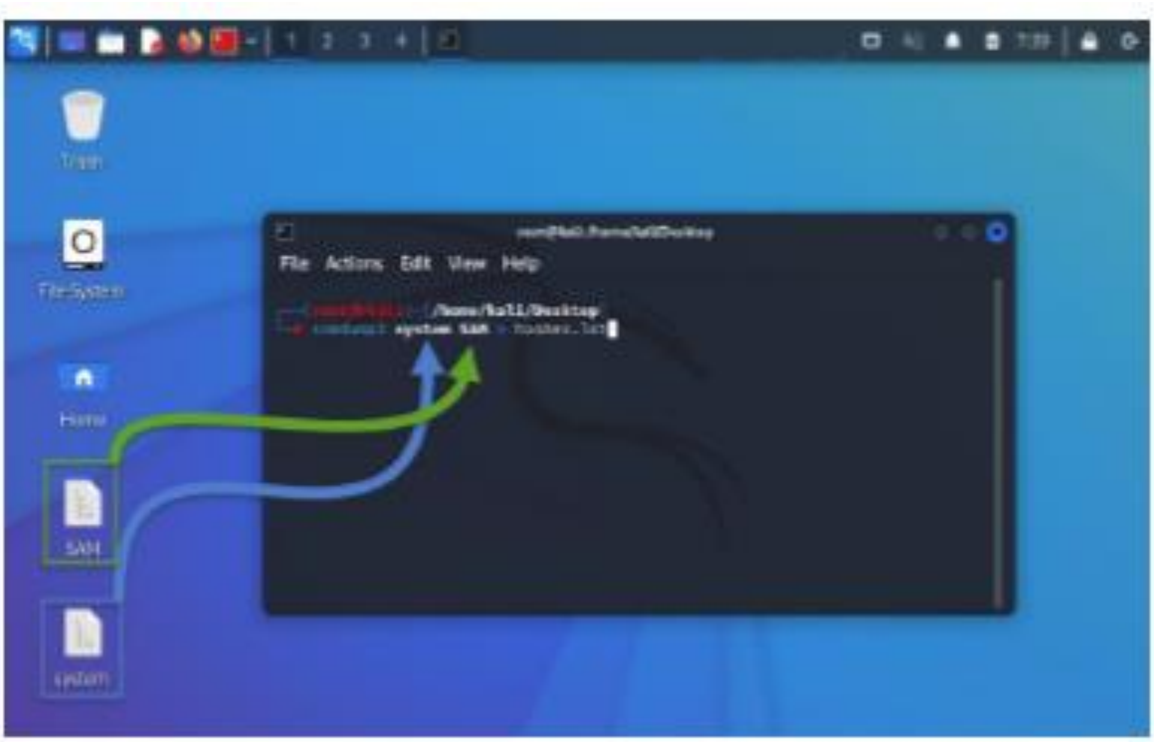
Type Command:
cd /home/kali/Desktop



Step 6: Run samdump2 to generate and save a list of user accounts (Windows) with NTLM hashes.

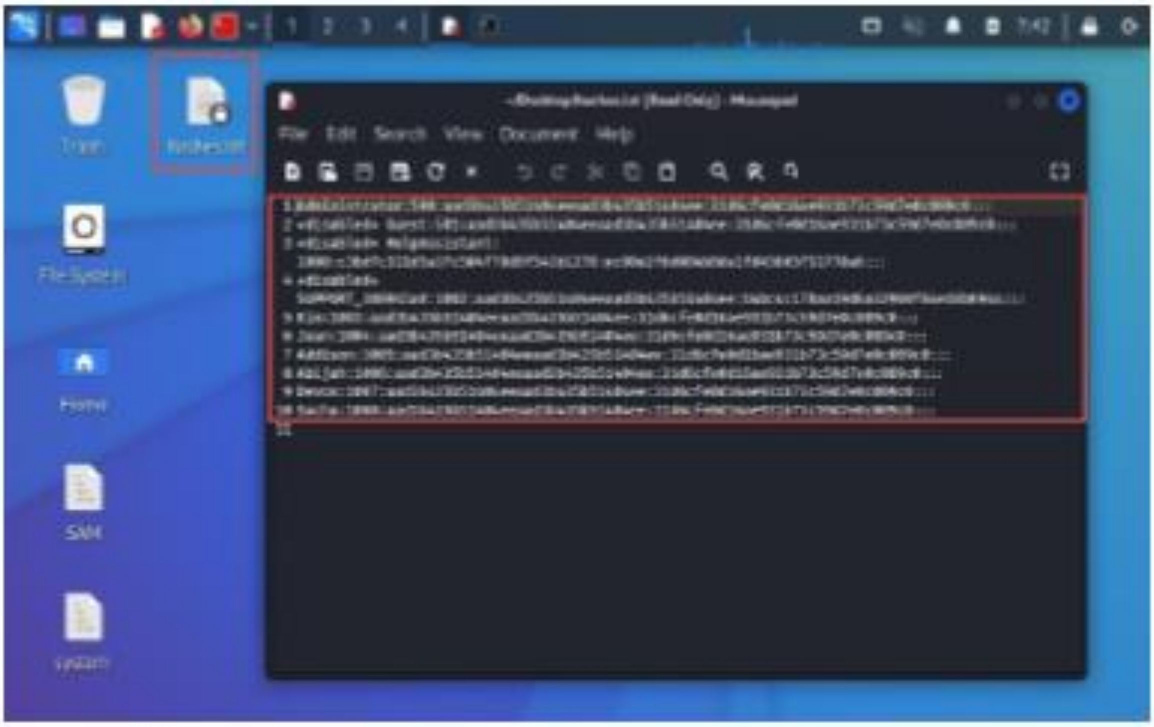
Command Syntax:
samdump2 <system_hive>
<sam_hive> > output_file.lst

Example:
Samdump2 system SAM > hashes.lst



Step 7: Open the created “hashes.lst” file to display User Accounts and Hashes.

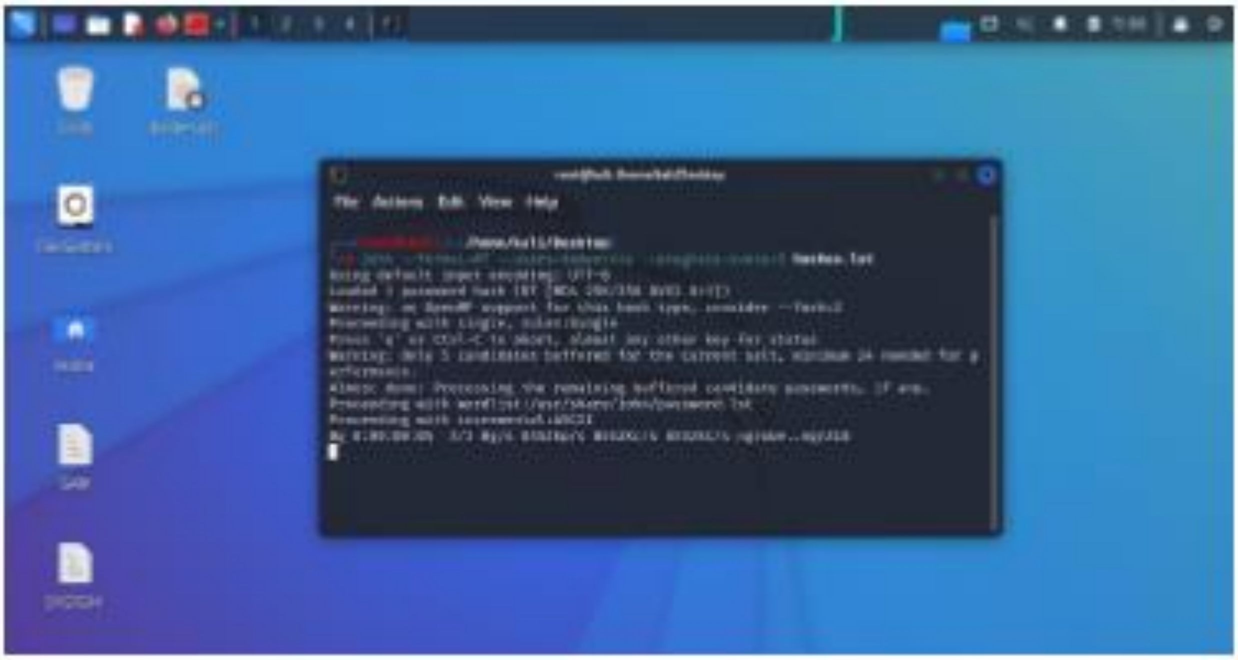
Hash Example/Description:
Jean (Username)
1003(SID)
aad3b435b51404eeaad3b435b51404ee (LM Hash)
1b3801b608a6be89d21fd3c5729d30bf (NTLM Hash)



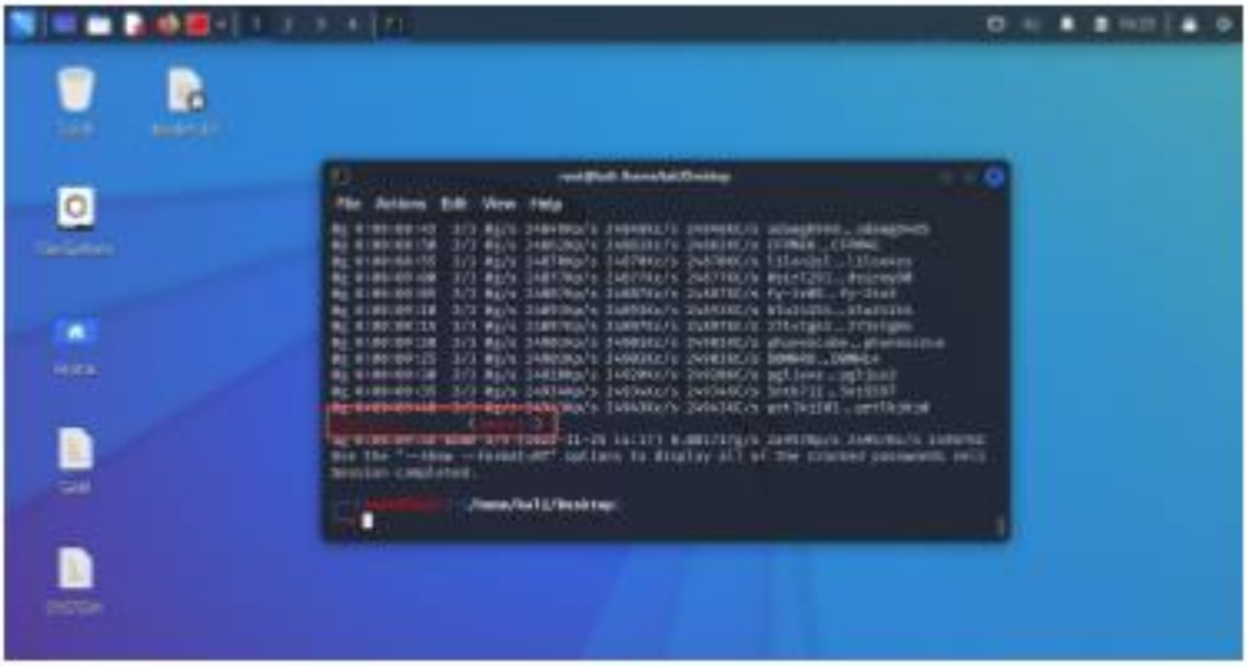
Step 8: Run John-the-Ripper program. Wait until password is cracked.

Command Syntax:
john --format=<format> --users=<account_name> --progress-every=5 --max-length=N <hashlist_file>

Example:
john --format=NT --users=Jean --progress-every=5 --maxlength=10 hashes.lst



Patiently wait for the password be cracked.



LESSON 4

Overview of Steganography

Learning Outcomes:

- Define steganography and distinguish it from cryptography.
- Describe common steganographic techniques used to hide data in images, audio, and video files.
- Demonstrate the use of basic steganography tools for concealing and extracting hidden messages.

Understanding Steganography

The word **steganography**—the art of secret writing—is derived from the Greek words steganos, meaning "covered," and graphein, meaning "to write." The Greek historian Herodotus described one of the first steganographers, a fellow Greek who warned of an imminent invasion by writing a message on the wood beneath a wax writing tablet.8 While steganography is technically not a form of cryptography, it is another way of protecting the confidentiality of information in transit. The most popular modern version of steganography involves hiding information within files that contain digital pictures or other images.

Some steganographic tools can calculate the largest image that can be stored before being detectable. Messages can also be hidden in computer files that do not hold images; if such files do not use all of their available bits, data can be placed where software ignores it and people almost never look.

Some applications can hide messages in .bmp, .wav, .mp3, and .au files, as well as in otherwise unused storage space on CDs and DVDs. Another approach is to hide a message in a text or document file and store the payload in what appears to be unused whitespace.

History of Steganography

Steganography is the practice of concealing a secret message behind a normal message. It stems from two Greek words, which are steganos, means covered and graphia, means writing. Steganography is an ancient practice, being practiced in various forms for thousands of years to keep communications private.

For Example:

- The first use of steganography can be traced back to 440 BC when ancient Greece, people wrote messages on wood and covered it with wax, that acted as a covering medium
- Romans used various forms of Invisible Inks, to decipher those hidden messages light or heat were used
- During World War II the Germans introduced microdots, which were complete documents, pictures, and plans reduced in size to the size of a dot and were attached to normal paperwork
- Null Ciphers were also used to hide unencrypted secret messages in an innocent looking normal message

Now, we have a lot of modern steganographic techniques and tools to make sure that knows our data remains secret. Now you might be wondering if steganography is same as cryptography. No, they are two different concepts and this steganography tutorial presents you the main differences between them.

Invisible Ink

Wax Tablet

Microdots

Null Cipher

	STEGANOGRAPHY	CRYPTOGRAPHY
Definition	It is a technique to hide the existence of communication	It's a technique to convert data into an incomprehensible form
Purpose	Keep communication secure	Provide data protection
Data Visibility	Never	Always
Data Structure	Doesn't alter the overall structure of data	Alters the overall structure of data
Key	Optional, but offers more security if used	Necessary requirement
Failure	Once the presence of a secret message is discovered, anyone can use the secret data	If you possess the decryption key, then you can figure out original message from the ciphertext

At their core, both of them have almost the same goal, which is protecting a message or information from the third parties. However, they use a totally different mechanism to protect the information.

Cryptography changes the information to ciphertext which cannot be understood without a decryption key. So, if someone were to intercept this encrypted message, they could easily see that some form of encryption had been applied. On the other hand, steganography does not change the format of the information but it conceals the existence of the message.

Steganography Techniques

Depending on the nature of the cover object(actual object in which secret data is embedded), steganography can be divided into five types:

- Text Steganography
- Image Steganography
- Video Steganography
- Audio Steganography
- Network Steganography

Text Steganography

Hiding information inside the text files. It involves things like changing the format of existing text, changing words within a text, generating random character sequences or using context-free grammars to generate readable texts. Various techniques used to hide the data in the text are:

- Format Based Method
- Random and Statistical Generation
- Linguistic Method

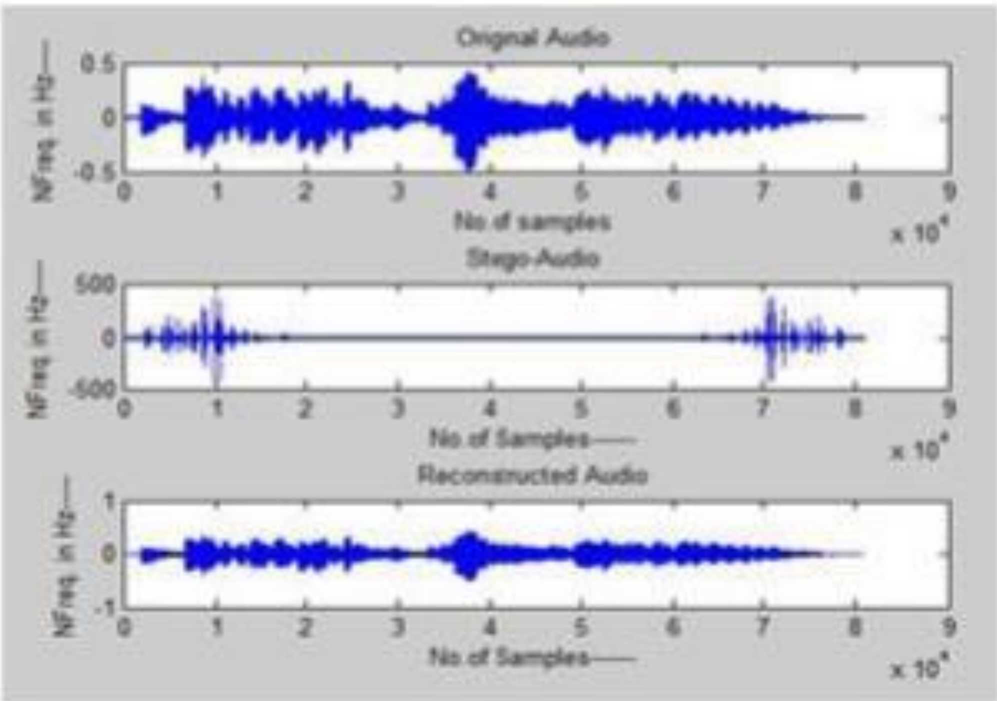
Image Steganography

Hiding the data by taking the cover object as the image is known as image steganography. In digital steganography, images are widely used cover source because there are a huge number of bits present in the digital representation of an image. There are a lot of ways to hide information inside an image. Common approaches include:

- Least Significant Bit Insertion
- Masking and Filtering
- Redundant Pattern Encoding
- Encrypt and Scatter
- Coding and Cosine Transformation

Audio Steganography
In audio steganography, the secret message is embedded into an audio signal which alters the binary sequence of the corresponding audio file. Hiding secret messages in digital sound is a much more difficult process when compared to others, such as Image Steganography. Different methods of audio steganography include:

- Least Significant Bit Encoding
- Parity Encoding
- Phase Coding
- Spread Spectrum



Video Steganography
In Video Steganography you can hide kind of data into digital video format. The advantage of this type is a large amount of data can be hidden inside and the fact that it is a moving stream of images and sounds. You can think of this as the combination of Image Steganography and Audio Steganography. Two main classes of Video Steganography include:

- Embedding data in uncompressed raw video and compressing it later
- Embedding data directly into the compressed data stream

Network Steganography (Protocol Steganography)
It is the technique of embedding information within network control protocols used in data transmission such TCP, UDP, ICMP etc. You can use steganography in some covert channels that you can find in the OSI model. For Example, you can hide information in the header of a TCP/IP packet in some fields that are either optional.

Steganography Techniques

These are the steganography tools which are available for free:

- **Stegosuite** is a free steganography tool which is written in Java. With Stegosuite you can easily hide confidential information in image files.
- **Steghide** is an open source Steganography software that lets you hide a secret file in image or audio file.

- **Xiao Steganography** is a free software that can be used to hide data in BMP images or in WAV files.
- **SSuite Picsel** is another free portable application to hide text inside an image file but it takes a different approach when compared to other tools.
- **OpenPuff** is a professional steganographic tool where you can store files in image, audio, video or flash files

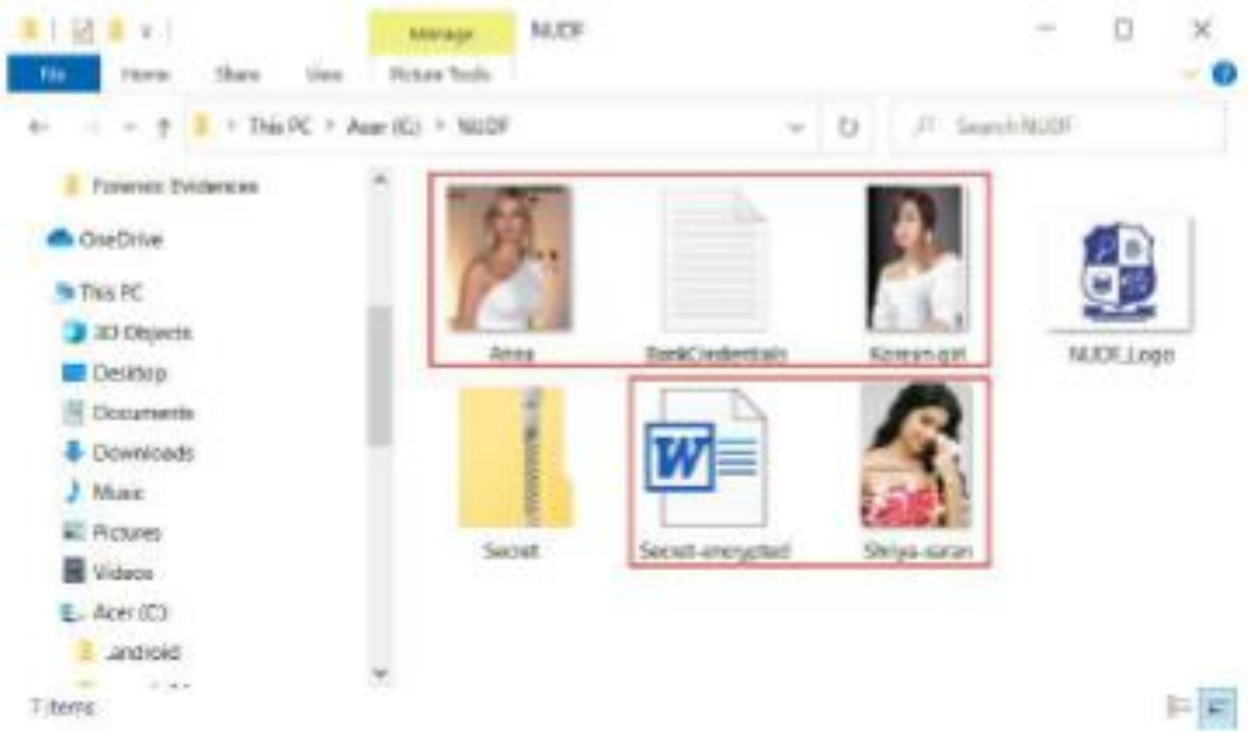
Best Tools to Detect Steganography

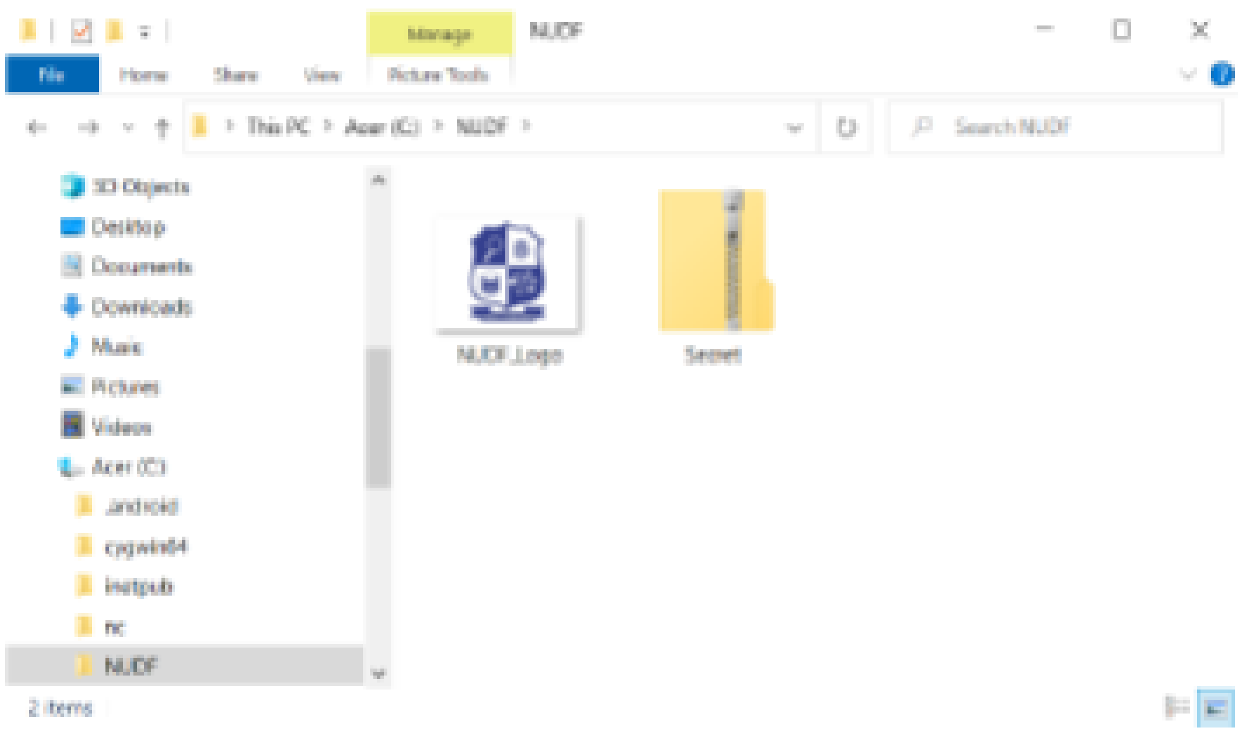
These are the steganography tools which are available for free:
- **StegDetect** (JPG/BMP/GIF Files) is another simple tool to detect steganography in images. Just like other tools, it detects the message in images. But it uses different approach for that. Here it takes an image and then marks the corresponding pixels in in that image that may contains the hidden message. This is actually a developer tool and to run this, you will need Python on your PC. After analyzing an image, it saves a part of the image that may contain the hidden message.
- **StegExpose** (PNG/BMP Files) is a simple command line utility that you can use to detect steganography in images. This is a cross-platform tool that you can use easily use to detect hidden messages in all the images of a specific directory. It scans every image in the specified directory and can tell you if there is a hidden message in it along with its estimated size. It uses a powerful algorithm to detect steganography in images and show you the result. Just like the software above, it is an open source and a dedicated tool to detect steganography in images.

Practical Steganography and Detection: Basic Steganography

Use Image File (JPEG) to Hide Files

1. Create a folder anywhere in your PC and keep all the files you want to hide as well as the jpg image you will use to keep that files in.
2. Just select all the files you want to hide, and by right clicking on them select the option of add them to a compressed ZIP or RAR file. Only compress the files you want to hide, not the jpg image. Lets name it as “Secret.rar”. You’ll see that there is those files, the jpg image and a compressed archive named Secret.rar inside the folder NUDF

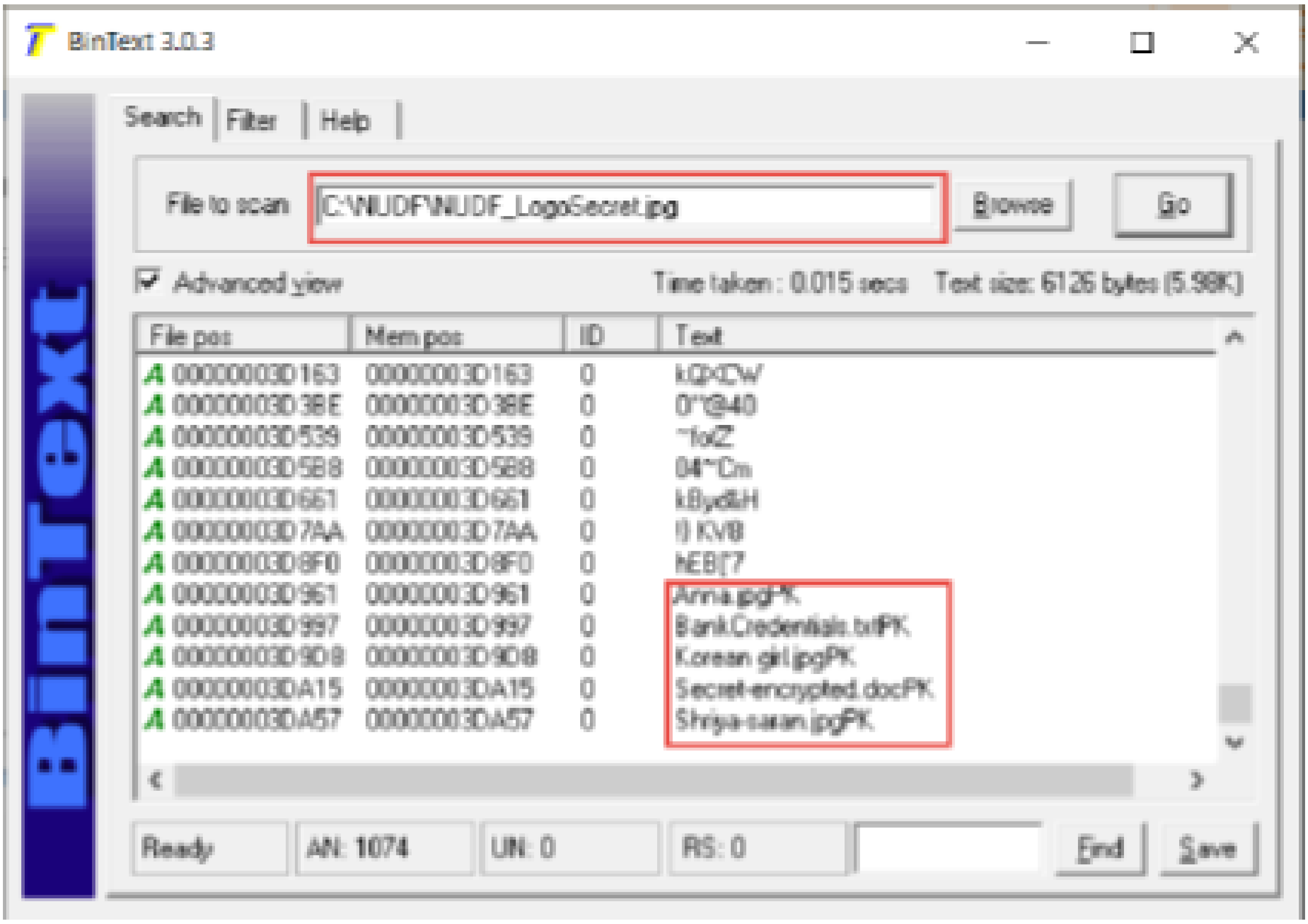




3. Click start menu, run, cmd. In Command Prompt type cd to get into the root directory and again type cd followed by the folder's name you want to hide i.e. "cd NUDF". Now you are inside the directory C:\NUDF>. Now type in copy /b picturename.jpg + foldername.zip outputfilename.jpg

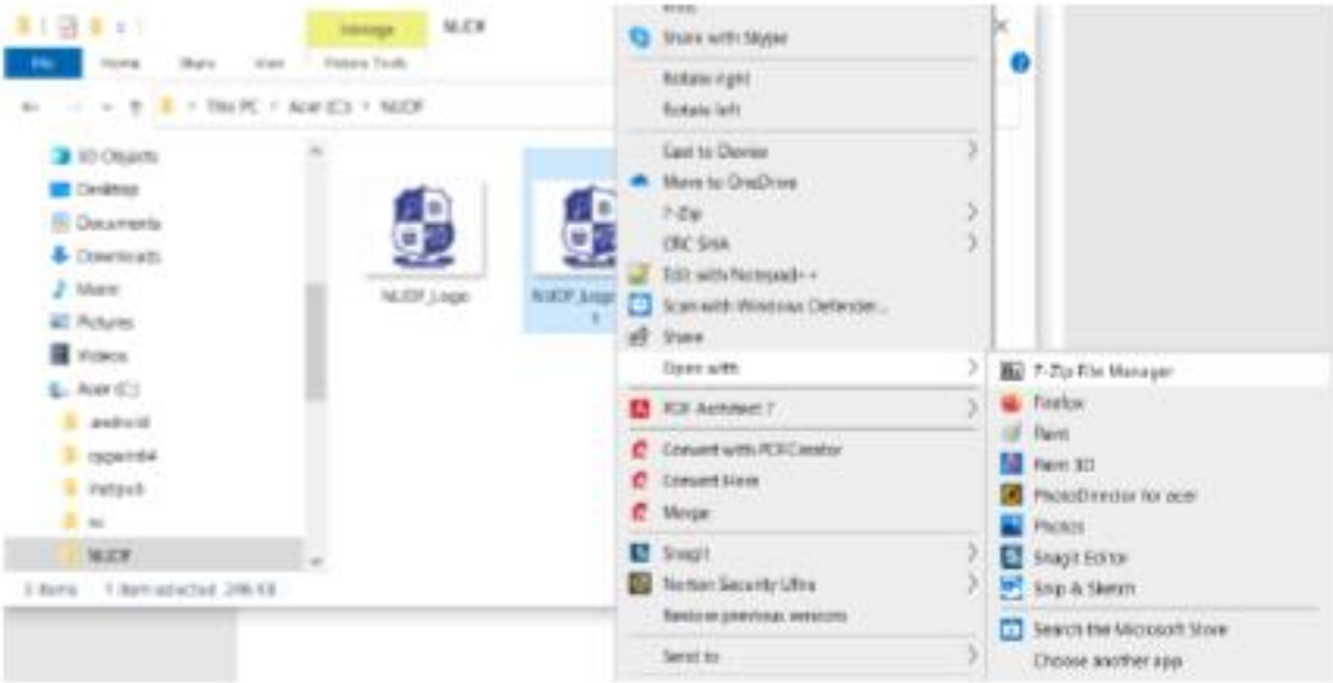
Detection Mechanism

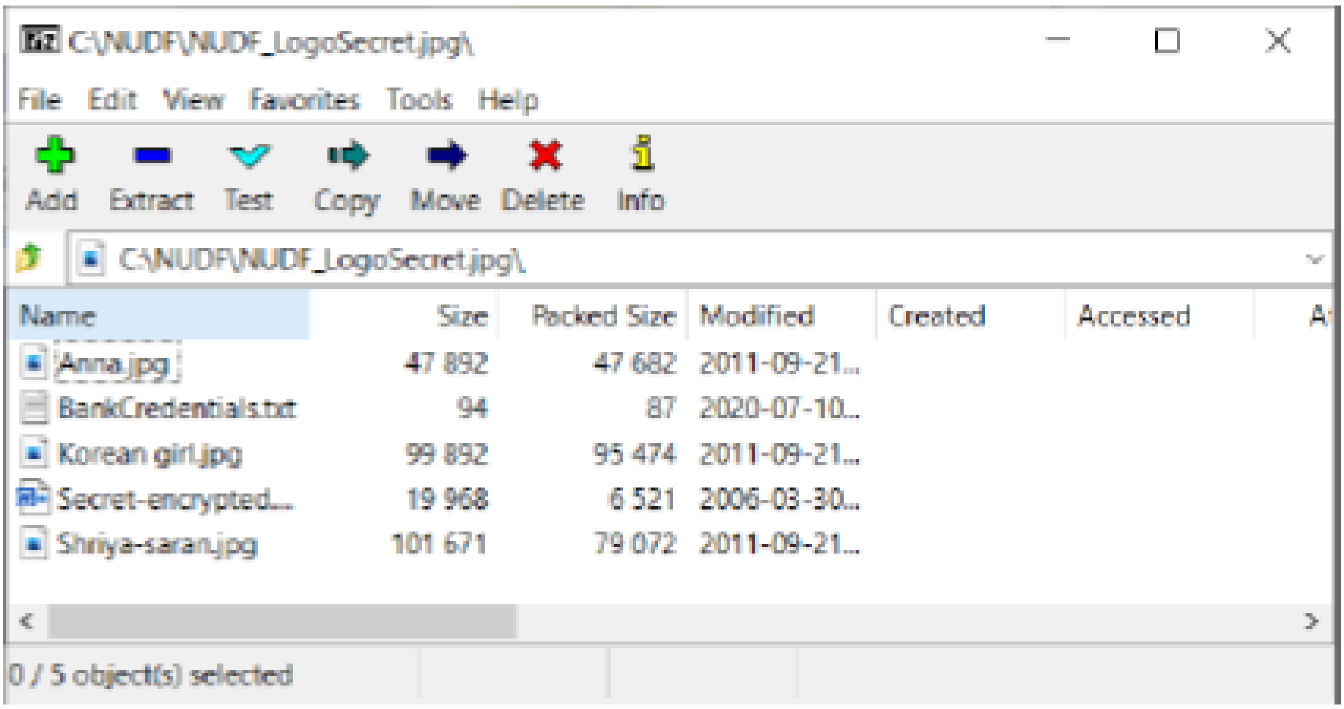
To uncover the hidden files, you may use BinText to search strings or investigate the string content of a file.



Accessing the Hidden File

Just right click on the file and chose open with WinRar. Then you can see the file you hide behind the picture. Another way to access the hidden file is that just change the extension of the file to .RAR and open the file using WinRar and then extract them.

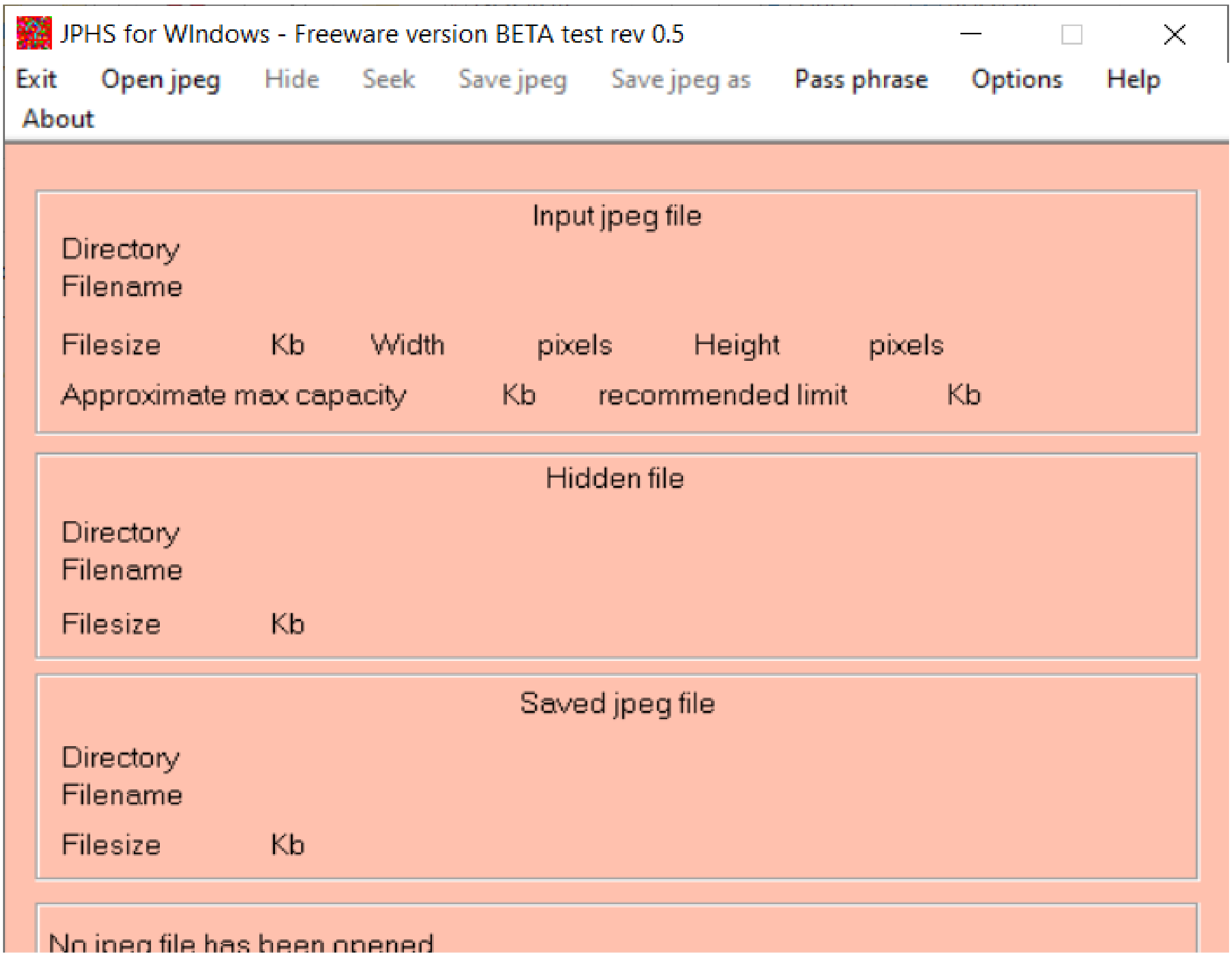




Practical Steganography and Detection: Advance Steganography

Use JPHide (Image Steganography)

JPHIDE and **JPSEEK** are programs which allow you to hide a file in a jpeg visual image. There are lots of versions of similar programs available on the internet, but JPHIDE and JPSEEK are rather special. The design objective was not simply to hide a file but rather to do this in such a way that it is impossible to prove that the host file contains a hidden file.

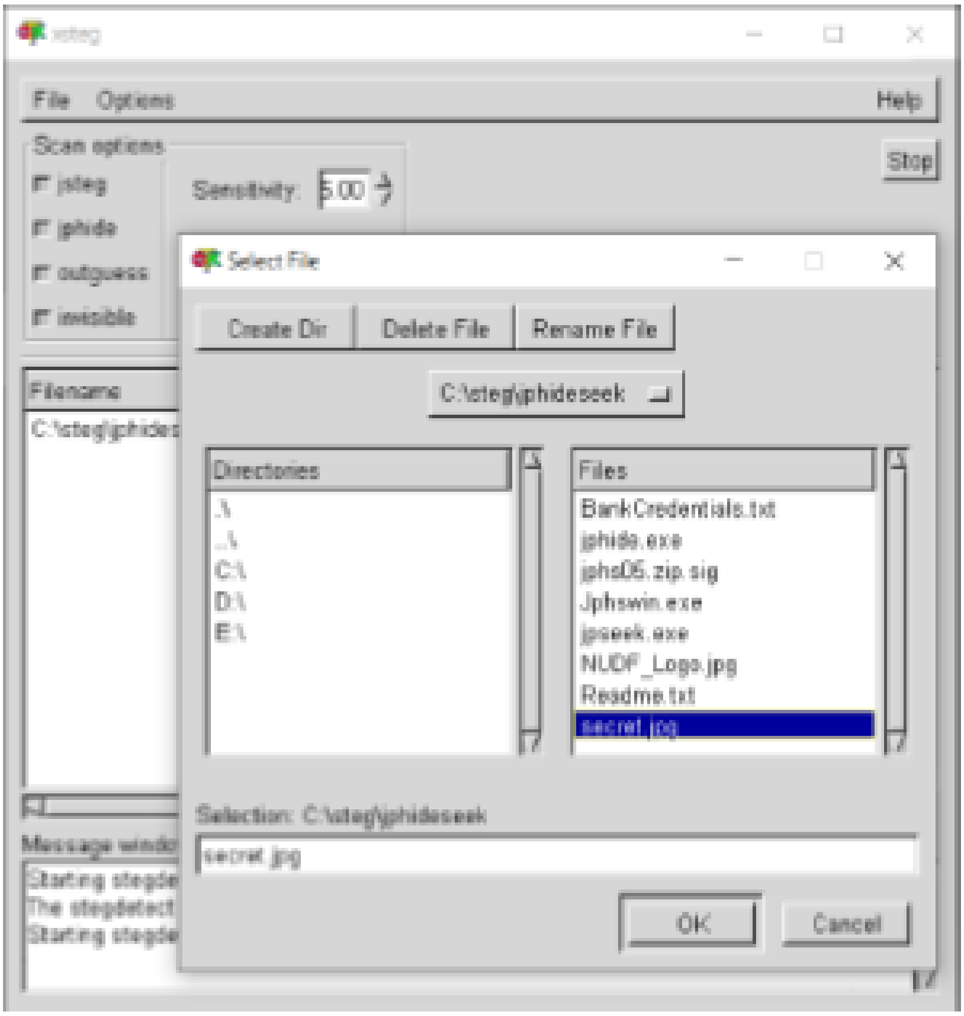


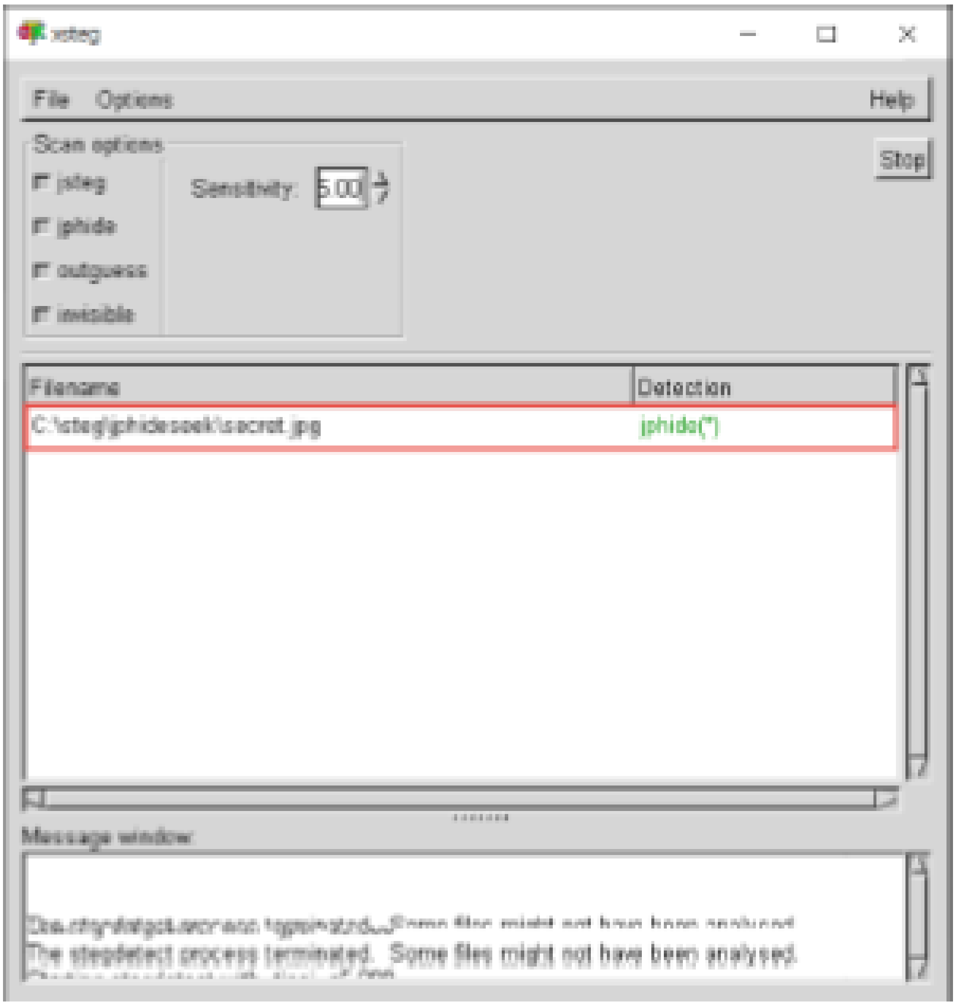
Detection Mechanism using StegDetect/xSteg

The **stegdetect** utility analyses image files for steganographic content. It runs statistical tests to determine if steganographic content is present, and also tries to find the system that has been used to embed the hidden information.

The options are as follows:

- -q Only reports images that are likely to have steganographic content.
- -s float Changes the sensitivity of the detection algorithms. Their results are multiplied by the specified number. The higher the number the more sensitive the test will become. The default is 1.
- -t tests Sets the tests that are being run on the image. The following characters are understood:
    - j Tests if information has been embedded with jsteg.
    - o Tests if information has been embedded with outguess.
    - p Tests if information has been embedded with jphide.
    - i Tests if information has been hidden with invisible secrets.


xsteg
File
Options
Help
Scan options
jsteg
jphide
outguess
invisible
Sensitivity:
5.00
Stop
Select File
Create Dir
Delete File
Rename File
C:\steg\jphideseek
Directories
.\
..\
C:\
D:\
E:\
Files
BankCredentials.txt
jphide.exe
jphs05.zip.sig
Jphswin.exe
jpseek.exe
NUDF_Logo.jpg
Readme.txt
secret.jpg
Selection: C:\steg\jphideseek
secret.jpg
OK
Cancel
Filename
Message window


xsteg
File
Options
Help
Scan options
jsteg
jphide
outguess
invisible
Sensitivity:
5.00
Stop
Filename
Detection
C:\steg\jphideseek\secret.jpg
jphide(*)
Message window
The stegdetect process terminated. Some files might not have been analysed.

SEATWORK ACTIVITY

Cryptography and Steganography

Using various tools, be able to perform "Cipher Decoding, Cryptographic Attack, and Steganography Detection and Extraction".

Disclaimer: Activity is for educational purposes only! Misuse or targeting other services outside the controlled or virtual environment is punishable by law and the University. The University and the Instructor have no liability for misuse of the tools used in this exercise.

General Procedure(s)

Task 1 Cipher Decoding and Cryptographic Attack

Using cryptography concepts, techniques and tools, be able to perform the following:

- Cipher decoding, also known as decryption or deciphering, is the process of converting ciphertext back into its original plaintext form using a specific key and algorithm.
- Cryptographic attacks are malicious attempts to bypass the security of cryptographic systems by exploiting weaknesses in algorithms, keys, or implementations. These attacks aim to gain unauthorized access to sensitive information, compromising confidentiality, integrity, and availability.

Task 2 Steganography Detection and Extraction

Using steganography concepts, techniques and tools, be able to perform the following:

- Steganography detection, also known as steganalysis, involves identifying hidden data embedded within seemingly innocuous digital media like images, audio, or video.
- Steganography extraction is the process of recovering hidden information from a file or message that has been modified to contain it. This involves identifying the hidden data within the seemingly ordinary content and extracting it, often using specific algorithms or tools based on the steganography method employed.

Note: Refer to the course module lesson as a guide.

Show proof of your activity completion by providing clear screenshot of your entire desktop environment and/or actual photos (clear) based on the tasks performed. Use the activity template in providing the necessary screenshots.

LESSON SUMMARY

Encryption is the process of converting a message into a form that is unreadable to unauthorized people.

The science of encryption, known as cryptology, encompasses cryptography (making and using encryption codes) and cryptanalysis (breaking encryption codes).

Two basic processing methods are used to convert plaintext data into encrypted data—bit stream and block ciphering. The other major methods used for scrambling data include substitution ciphers, transposition ciphers, the XOR function, the Vigenère cipher, and the Vernam cipher.

The strength of many encryption applications and cryptosystems is determined by key size. All other things being equal, the length of the key directly affects the strength of the encryption.

Hash functions are mathematical algorithms that generate a message summary, or digest, that can be used to confirm the identity of a specific message and confirm that the message has not been altered.

Most cryptographic algorithms can be grouped into two broad categories: symmetric and asymmetric. In practice, most popular cryptosystems are hybrids that combine symmetric and asymmetric algorithms.

A cryptographic attack is a method used by hackers to target cryptographic solutions like ciphertext, encryption keys, etc. These attacks aim to retrieve the plaintext from the ciphertext or decode the encrypted data. Hackers may attempt to bypass the security of a cryptographic system by discovering weaknesses and flaws in cryptography techniques, cryptographic protocol, encryption algorithms, or key management strategy.

Steganography is the hiding of information. It is not properly a form of cryptography but is similar in that it is used to protect confidential information while in transit.

KEY TERMS

- Cryptanalysis
- Cryptography
- Cryptology
- Decipher
- Encipher
- Ciphertext
- Plaintext
- Key
- Cryptovariable
- Substitution Cipher
- Vigenère Cipher
- Monoalphabetic Substitution
- Polyalphabetic Substitution
- Transposition Cipher
- Permutation Cipher
- Hash Function
- Hash Algorithms
- Private-key Encryption
- Symmetric Encryption
- Asymmetric Encryption
- Public-key Encryption
- Digital Certificates
- Digital Signatures
- Session Keys
- Steganography
- Text Steganography
- Image Steganography
- Video Steganography
- Audio Steganography
- Network Steganography

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